

《数据库系统原理》实验报告（实验二）					
题目：DML 语言实验					
学号	2351520	姓名	毛星博	日期	2025.4.15
实验环境：Docker OceanBase					
实验步骤及结果截图： 0. 建立数据库 lib_borrow 并使用 <pre>obclient(root@sys)[(none)]> create database lib_borrow; Query OK, 1 row affected (0.047 sec) obclient(root@sys)[(none)]> use lib_borrow; Database changed</pre> 1. 在数据库建立相关数据表 <pre>obclient(root@sys)[lib_borrow]> create table category(-> no int primary key not null, -> name varchar(100) not null ->); Query OK, 0 rows affected (0.066 sec) obclient(root@sys)[lib_borrow]> create table books(-> no int primary key not null, -> title varchar(100) not null, -> author varchar(100) not null, -> publication_year int not null, -> c_no int not null, -> foreign key (c_no) references category(no) ->); Query OK, 0 rows affected (0.074 sec) obclient(root@sys)[lib_borrow]> create table readers (-> no integer primary key not null, -> name varchar(50) not null, -> gender varchar(6) not null check (gender in ('Female', 'Male')), -> tel varchar(20) not null ->); Query OK, 0 rows affected (0.040 sec) obclient(root@sys)[lib_borrow]> create table borrow_records (-> no integer primary key not null, -> b_no integer not null, -> r_no integer not null, -> borrow_date date not null, -> return_date date, -> foreign key (b_no) references books(no), -> foreign key (r_no) references readers(no) ->); Query OK, 0 rows affected (0.054 sec)</pre>					
查看表结构					

```
obclient(root@sys)[lib_borrow]> desc books;
```

Field	Type	Null	Key	Default	Extra
no	int(11)	NO	PRI	NULL	
title	varchar(100)	NO		NULL	
author	varchar(100)	NO		NULL	
publication_year	int(11)	NO		NULL	
c_no	int(11)	NO		NULL	

```
5 rows in set (0.001 sec)
```

```
obclient(root@sys)[lib_borrow]> desc category;
```

Field	Type	Null	Key	Default	Extra
no	int(11)	NO	PRI	NULL	
name	varchar(100)	NO		NULL	

```
2 rows in set (0.001 sec)
```

```
obclient(root@sys)[lib_borrow]> desc readers;
```

Field	Type	Null	Key	Default	Extra
no	int(11)	NO	PRI	NULL	
name	varchar(50)	NO		NULL	
gender	varchar(6)	NO		NULL	
tel	varchar(20)	NO		NULL	

```
4 rows in set (0.004 sec)
```

```
obclient(root@sys)[lib_borrow]> desc borrow_records;
```

Field	Type	Null	Key	Default	Extra
no	int(11)	NO	PRI	NULL	
b_no	int(11)	NO		NULL	
r_no	int(11)	NO		NULL	
borrow_date	date	NO		NULL	
return_date	date	YES		NULL	

```
5 rows in set (0.004 sec)
```

插入具体数据

```
obclient(root@sys)[lib_borrow]> insert into category (no, name) values
```

```
-> (1, '计算机'),
-> (2, '文学'),
-> (3, '历史'),
-> (4, '科学');
```

```
ERROR 1062 (23000): Duplicate entry '1' for key 'PRIMARY'
```

```
[172.17.0.2:2882] [2025-04-15 07:51:32.569206] [YB42AC110002-000632CC1A11618F-0-0]
```

```
obclient(root@sys)[lib_borrow]> -- 插入books表数据
```

```
Query OK, 0 rows affected (0.000 sec)
```

```
obclient(root@sys)[lib_borrow]> insert into books (no, title, author, publication_year, c_no) values
```

```
-> (1, 'Python编程实战', '张三', 2021, 1),
-> (2, '百年孤独', '加西亚·马尔克斯', 1967, 2),
-> (3, '明朝那些事儿', '当年明月', 2006, 3),
-> (4, '时间简史', '霍金', 1988, 4),
-> (5, 'Java核心技术', '凯S.霍斯特曼', 2019, 1);
```

```
Query OK, 5 rows affected (0.010 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
obclient(root@sys)[lib_borrow]> insert into readers (no, name, gender, tel) values
```

```
-> (1, '小明', 'Male', '13800138000'),
-> (2, '小红', 'Female', '13900139000'),
-> (3, '小刚', 'Male', '13600136000'),
-> (4, '小美', 'Female', '13700137000'),
-> (5, '小强', 'Male', '13300133000');
```

```
Query OK, 5 rows affected (0.008 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
obclient(root@sys)[lib_borrow]> insert into borrow_records (no, b_no, r_no, borrow_date, return_date) values
```

```
-> (1, 1, 1, '2024-10-01', '2024-10-20'),
-> (2, 2, 2, '2024-10-05', '2024-10-25'),
-> (3, 1, 3, '2024-10-10', '2024-10-30'),
-> (4, 3, 4, '2024-10-15', '2024-11-01'),
-> (5, 2, 5, '2024-10-20', null);
```

```
Query OK, 5 rows affected (0.008 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

2. 使用比较运算符查询 2020 年以后出版的图书的 Title 和 publication_year。

```
obclient(root@sys)[lib_borrow]> select title,publication_year
-> from books
-> where publication_year >= 2020;
+-----+-----+
| title          | publication_year |
+-----+-----+
| Python编程实战 | 2021 |
+-----+-----+
1 row in set (0.002 sec)
```

3. 查询借阅过编号为 3 的图书的读者的 Name 和 Tel。

```
obclient(root@sys)[lib_borrow]> select r.name, r.tel
-> from readers r
-> join borrow_records br on r.no = br.r_no
-> where br.b_no = 3;
+-----+-----+
| name | tel |
+-----+-----+
| 小美 | 13700137000 |
+-----+-----+
1 row in set (0.009 sec)
```

4. 查询作者姓名中包含“张”字的图书信息。

```
obclient(root@sys)[lib_borrow]> select *
-> from books
-> where author like '%张%';
+-----+-----+-----+-----+-----+
| no | title          | author | publication_year | c_no |
+-----+-----+-----+-----+-----+
| 1 | Python编程实战 | 张三 | 2021 | 1 |
+-----+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

5. 查询所有男性读者的借阅记录，包括读者姓名、借阅图书 Title、借阅日期和归还日期，结果按借阅日期升序排列。

```
obclient(root@sys)[lib_borrow]> select r.name,b.title,br.borrow_date,br.return_date
-> from borrow_records as br
-> join readers as r on br.r_no = r.no
-> join books as b on br.r_no = b.no
-> where r.gender = 'Male'
-> order by br.borrow_date;
+-----+-----+-----+-----+
| name | title          | borrow_date | return_date |
+-----+-----+-----+-----+
| 小明 | Python编程实战 | 2024-10-01 | 2024-10-20 |
| 小刚 | 明朝那些事儿 | 2024-10-10 | 2024-10-30 |
| 小强 | Java核心技术 | 2024-10-20 | NULL |
+-----+-----+-----+-----+
3 rows in set (0.007 sec)
```

6. 查询女性读者的总人数。

```
obclient(root@sys)[lib_borrow]> select count(*) as female_num
-> from readers
-> where gender = 'Female';
+-----+
| female_num |
+-----+
| 2 |
+-----+
1 row in set (0.001 sec)
```

7. 查询借阅时长超过 15 天的借阅记录，包括读者姓名、图书 Title。

```
obclient(root@sys)[lib_borrow]> select r.name, b.title
-> from borrow_records br
-> join readers r on br.r_no = r.no
-> join books b on br.b_no = b.no
-> where br.return_date is not null and datediff(return_date,borrow_date)>15;
+-----+-----+
| name | title |
+-----+-----+
| 小刚 | Python编程实战 |
| 小红 | 百年孤独 |
| 小美 | 明朝那些事儿 |
| 小明 | Python编程实战 |
+-----+-----+
4 rows in set (0.002 sec)
```

8. 向 `borrow_records` 表中插入一条记录（借阅编号为 6，图书编号为 5，读者编号为 5，借阅日期为 '2025-03-24'，归还日期为空）。

```
obclient(root@sys)[lib_borrow]> insert into borrow_records (no,b_no,r_no,borrow_date,return_date)
-> values (6,5,5,'2025-03-24',null);
Query OK, 1 row affected (0.007 sec)
```

9. 计算每本图书的平均借阅时长，并输出平均借阅时长超过 15 天的图书的 No。

```
obclient(root@sys)[lib_borrow]> select b_no,avg(datediff(return_date,borrow_date)) as avg_borrow_days
-> from borrow_records
-> group by b_no
-> having avg(datediff(return_date,borrow_date))>15;
+-----+-----+
| b_no | avg_borrow_days |
+-----+-----+
| 1 | 19.5000 |
| 2 | 20.0000 |
| 3 | 17.0000 |
+-----+-----+
3 rows in set (0.002 sec)
```

10. 查询借阅过编号为 1 或者编号为 2 图书的读者 No。

```
obclient(root@sys)[lib_borrow]> select r_no
-> from borrow_records
-> where b_no in (1,2);
+-----+
| r_no |
+-----+
| 1 |
| 2 |
| 3 |
| 5 |
+-----+
4 rows in set (0.002 sec)
```

11. 查询既借阅过编号为 1 又借阅过编号为 3 图书的读者 No。

```
obclient(root@sys)[lib_borrow]> select r_no
-> from borrow_records
-> where b_no = 1
-> intersect
-> select r_no
-> from borrow_records
-> where b_no = 3;
Empty set (0.002 sec)
```

12. 建立一个包含图书名字，作者，出版年份和图书类别的视图（赋予列名为 `stitle,sauthor,spublicationyear,categoryname`）。【create view】

```
obclient(root@sys)[lib_borrow]> create view book_details as
-> select b.title as stitle,
-> b.author as sauthor,
-> b.publication_year as spublicationyear,
-> c.name as categoryname
-> from books b
-> join category c on b.c_no = c.no;
Query OK, 0 rows affected (0.077 sec)
```

```
obclient(root@sys)[lib_borrow]> select * from book_details;
+-----+-----+-----+-----+
| stitle          | sauthor          | spublicationyear | categoryname |
+-----+-----+-----+-----+
| Python编程实战 | 张三            | 2021 | 计算机 |
| 百年孤独       | 加西亚·马尔克斯 | 1967 | 文学   |
| 明朝那些事儿   | 当年明月        | 2006 | 历史   |
| 时间简史       | 霍金            | 1988 | 科学   |
| Java核心技术   | 凯S.霍斯特曼    | 2019 | 计算机 |
+-----+-----+-----+-----+
5 rows in set (0.012 sec)
```

出现的问题:

在 **date** 数据类型相减时, 如果月份不同, 简单的使用 “-” 符号会出现错误。如在本次实验第 9 题中, 如果显示平均节约时长, 会发现《明朝那些事儿》的借阅时长明显不对, **2024-11-1** 到 **2024-10-15** 应该是 **17** 天, 但结果显示为 **86** 天。

```
obclient(root@sys)[lib_borrow]> select b.no, b.title,
->      avg(br.return_date - br.borrow_date) as avg_borrow_days
-> from books b
-> join borrow_records br on b.no = br.b_no
-> where br.return_date is not null
-> group by b.no, b.title
-> having avg(br.return_date - br.borrow_date) > 15;
+-----+-----+-----+
| no  | title          | avg_borrow_days |
+-----+-----+-----+
| 1   | Python编程实战 | 19.5000 |
| 2   | 百年孤独       | 20.0000 |
| 3   | 明朝那些事儿   | 86.0000 |
+-----+-----+-----+
3 rows in set (0.005 sec)
```

解决方案:

使用 **datediff** 函数。使用方法: **datediff** (被减数, 减数)

```
obclient(root@sys)[lib_borrow]> select b_no,avg(datediff(return_date,borrow_date)) as avg_borrow_days
-> from borrow_records
-> group by b_no
-> having avg(datediff(return_date,borrow_date))>15;
+-----+-----+
| b_no | avg_borrow_days |
+-----+-----+
| 1    | 19.5000         |
| 2    | 20.0000         |
| 3    | 17.0000         |
+-----+-----+
3 rows in set (0.002 sec)
```